

Respiration Papers (Directory)

Marguerite Butler

2025-11-26

Copies of these papers are located in the **shared google drive**. > RespirationPapers <https://drive.google.com/drive/folders/1Jcnx6t1sskNkcT2oJUaCiJAIspl1HUusT>

Important resources

Weibel Taylor Hoppeler 1991. Symmorphosis: A testable hypothesis of structure-function relationship **EVERYONE READ**

Fish

Wilkie 2011. Lamprey respiratory physiology, osmoregulation, and ionoregulation.

Wood Eom 2021. The osmorepiratory compromise in the fish gill <https://doi.org/10.1016/j.cbpa.2021.110895>. Very interesting paper on control of oxygen flux as well as iono and osmoregulation (next section!) pdf is in shared drive.

Early Tetrapods & Amphibians & Misc

Alhberg Clack 2006. A firm step from water to land (early tetrapod evol)

Carroll 2001. Early tetrapod evolution.

Dudley 1998. Atmospheric oxygen, giant paleozoic insects and the evolution of aerial locomotor performance

Gans 1970. Respiration in early tetrapods: The Frog is a Red Herring

Schaeffer 1965. Rhipsidian (fish) - amphibian transition

Pierson 2009. The cardiopulmonary physiology of dinosaurs

Shelton Boutilier 1982. Apnea in amphibians and reptiles

Wang et al 2001. respiratory consequences of feeding in amphibians and reptiles

Reptiles

Bennett 1973. Ventilation in lizards: rest and activity

Crawford Kampe 1971. Lizard *Sauromalus obesus* physiological responses to temperature

Farmer and Carrier 2000. Respiration and gas exchange during recovery from exercise in the American alligator

Glass and Wood 1983. Respiration in reptiles

Gleeson and Bennett 1985. Respiratory and cardiovascular adjustments to exercise in reptiles

Milsom 1984. Breathing in gecko

Shelton Boutilier 1982. Apnea in amphibians and reptiles

Wang Carrier Hicks 1997. Ventilation and gas exchange in lizards during treadmill exercise

Wang et al 2001. respiratory consequences of feeding in amphibians and reptiles

Birds

Benzo, Sturke 1986. Sturkey's avian physiology – **Essential for anyone attempting a Bird model** Check out from library and See me for help

Berger, Hart, and Roy 1970. Respiration, oxygen consumption and heart rate in some birds during rest and exercise

Brakenberry, Avery, and Gleeson 1981. Respiration in exercising fowl

Brackenberry, Gleeson, and Avery 1982. Control of ventilation in running birds: effects of hypoxia, hyperoxia, and CO₂

Brackenberry, Darby, and El-Sayed 1989. Respiratory function in exercising fowl following occlusion of the thoracic air sacs

Butler 1991. (not me!) Exercise in birds

Calder Schmidt-Neilsen 1966. Evaporative cooling and respiratory alkalosis in the pigeon

Frappell et al 2001. Scaling of respiratory variables in birds

Hinds and Calder 1971. Tracheal Dead Space in the Respiration of Birds

Schmidt-Nielsen, Kanwiyher, Lasiewski, Cohn, Bretz 1969. Temperature regulation and respiration in the Ostrich

Mammals

Fortuna and Saibene 1991. Mechanics of breathing in horses at rest and during exercise

Gallivan McDonnell Forrest 1989. Comparative pulmonary mechanics in the horse and the cow

Hoppeler and Weibel 1998. Limits for oxygen and substrate transport in mammals **
more about symmorphosis **

Husczuk Et al 1990. Ventilatory control during exercise in calves with artificial hearts

Jones Longworth Lindholm Conley Karas Kayar Taylor 1989. O₂ transport in large mammals (horse vs steer)

Martin Hutchinson. 1979 Ventilation in two amphibians

[[Noel et al 2022](#)] The origin of the allometric scaling of lung ventilation in mammals. *Modeling including BMR, FMR (DMR), and MMR.*

Stahl 1967. Scaling of Respiratory Variables in Mammals

Weibel 1987. Scaling of structural and functional variables in the respiratory system (mammals)

Withers 1992. p 627-631: pages from your text with a couple of highlighted comments inserted by me.

[[Young et al 1992](#)] Some properties of the mammalian locomotory and respiratory systems in relation to body mass. *Coupling of locomotion and respiration at the trot-gallop transition.*

[[Back to the Design 3 Assignment](#)]

[[Shared google drive](#)]